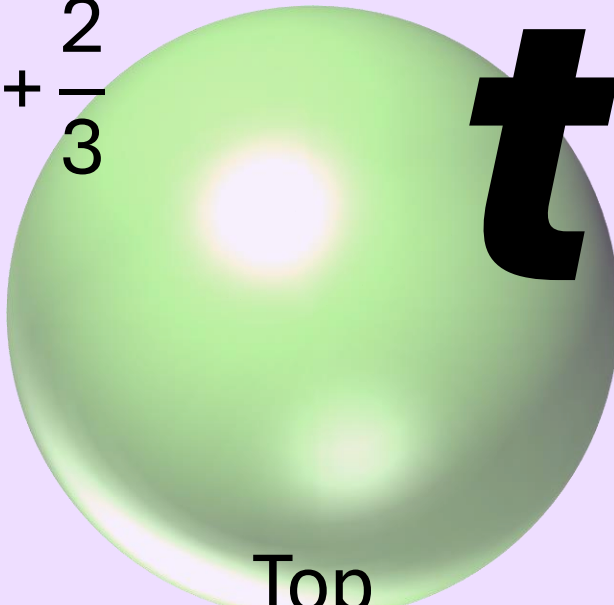
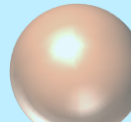
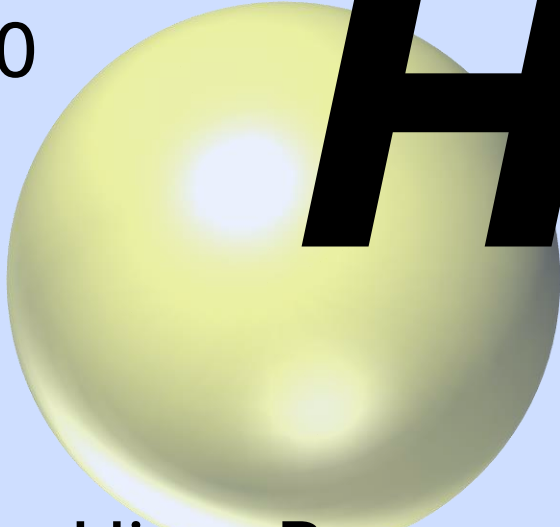
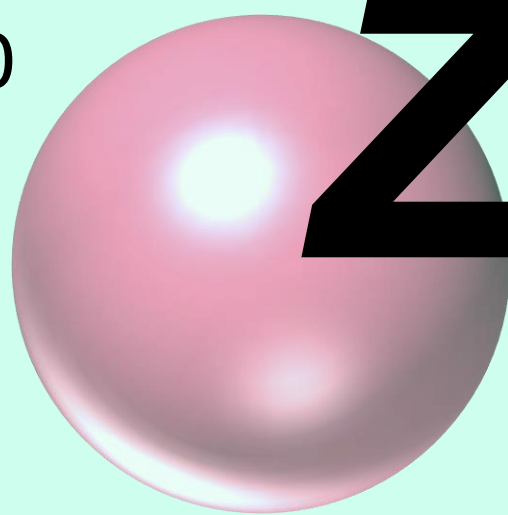


Fermions

Bosons

$+\frac{2}{3}$  <i>u</i> Up	$-\frac{1}{3}$  <i>d</i> Down	-1  <i>e</i> Electron	0 <i>ν_e</i> Electron Neutrino
$+\frac{2}{3}$  <i>c</i> Charm	$-\frac{1}{3}$  <i>s</i> Strange	-1  <i>μ</i> Muon	0 <i>ν_μ</i> Muon Neutrino
$+\frac{2}{3}$  <i>t</i> Top	$-\frac{1}{3}$  <i>b</i> Bottom	-1  <i>τ</i> Tau	0 <i>ν_τ</i> Tau Neutrino

0 <i>γ</i> Photon	0 <i>g</i> Gluon	0  <i>H</i> Higgs Boson
0  <i>Z</i> Z Boson	± 1  <i>W</i> W Boson	

Quarks

Leptons

Vector Bosons

Scalar Boson

Fermions

$+\frac{2}{3}$

***u***

Up

$-\frac{1}{3}$

***d***

Down

-1

***e***

Electron

0

ν_e

Electron Neutrino

$+\frac{2}{3}$

***c***

Charm

$-\frac{1}{3}$

***s***

Strange

-1

 **μ**

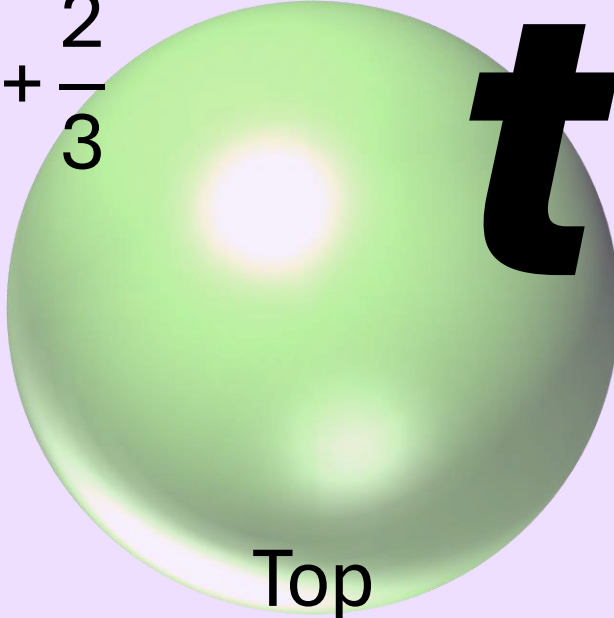
Muon

0

ν_μ

Muon Neutrino

$+\frac{2}{3}$

***t***

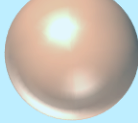
Top

$-\frac{1}{3}$

***b***

Bottom

-1

 **τ**

Tau

0

ν_τ

Tau Neutrino

Quarks

Leptons

Bosons

0

γ

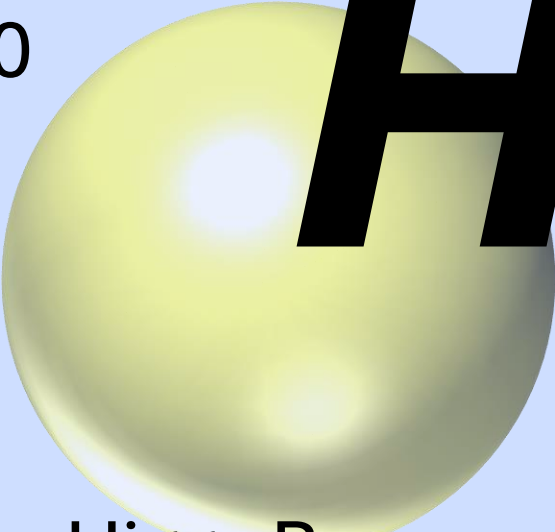
Photon

0

g

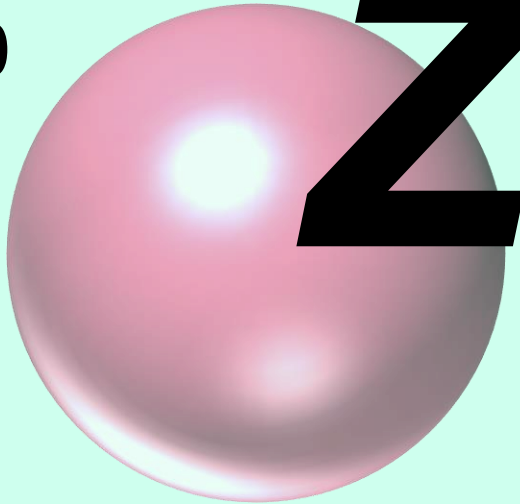
Gluon

0

***H***

Higgs Boson

0

***Z***

Z Boson

± 1

***W***

W Boson

Scalar
Boson

Vector Bosons

Fermions

$+\frac{2}{3}$

u
Up

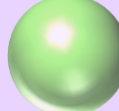
$-\frac{1}{3}$

d
Down

-1

e
Electron

0
 ν_e
Electron Neutrino

$+\frac{2}{3}$

c
Charm

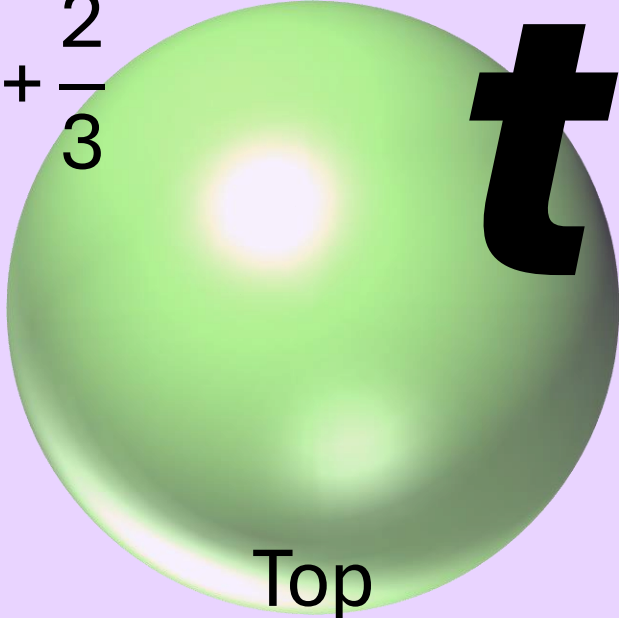
$-\frac{1}{3}$

s
Strange

-1

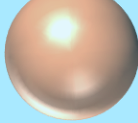
 μ
Muon

0
 ν_μ
Muon Neutrino

$+\frac{2}{3}$

t
Top

$-\frac{1}{3}$

b
Bottom

-1

 τ
Tau

0
 ν_τ
Tau Neutrino

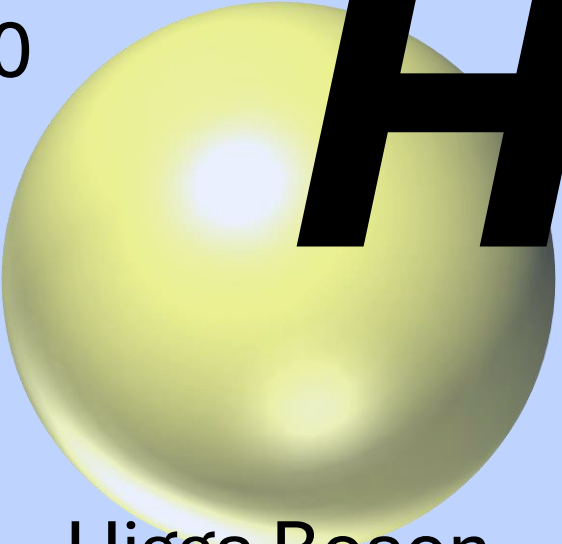
Quarks

Leptons

Bosons

0
γ
Photon

0
g
Gluon

0

H
Higgs Boson

0

Z
Z Boson

± 1

W
W Boson

Scalar
Boson

Vector Bosons

Fermions

$+\frac{2}{3}$

u
Up

$-\frac{1}{3}$

d
Down

-1

e
Electron

0
 ν_e
Electron Neutrino

$+\frac{2}{3}$

c
Charm

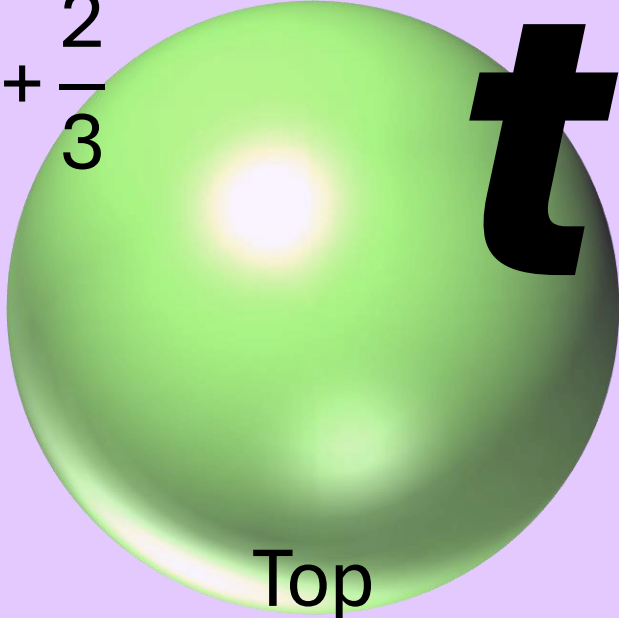
$-\frac{1}{3}$

s
Strange

-1

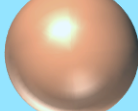
 μ
Muon

0
 ν_μ
Muon Neutrino

$+\frac{2}{3}$

t
Top

$-\frac{1}{3}$

b
Bottom

-1

 τ
Tau

0
 ν_τ
Tau Neutrino

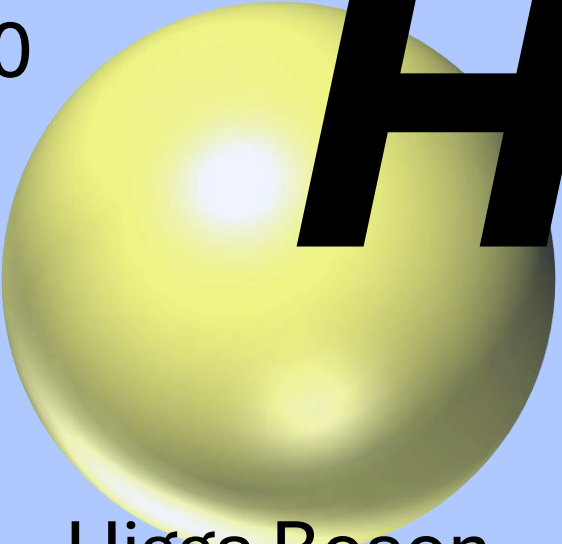
Quarks

Leptons

Bosons

0
 γ
Photon

0
g
Gluon

0

H
Higgs Boson

0

Z
Z Boson

± 1

W
W Boson

Vector Bosons

Scalar
Boson

Fermions

$+\frac{2}{3}$

u
Up

$-\frac{1}{3}$

d
Down

-1

e
Electron

0
 ν_e
Electron Neutrino

$+\frac{2}{3}$

c
Charm

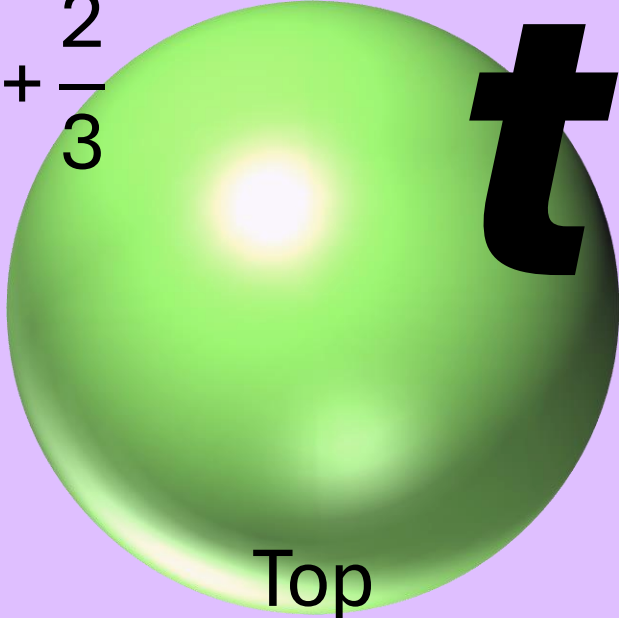
$-\frac{1}{3}$

s
Strange

-1

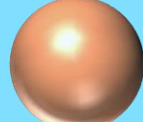
 μ
Muon

0
 ν_μ
Muon Neutrino

$+\frac{2}{3}$

t
Top

$-\frac{1}{3}$

b
Bottom

-1

 τ
Tau

0
 ν_τ
Tau Neutrino

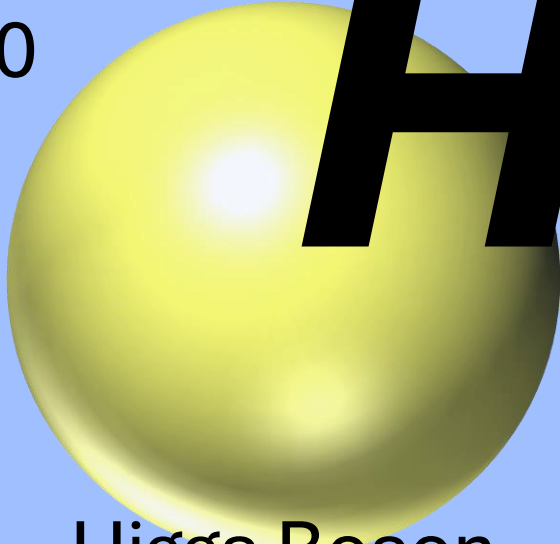
Quarks

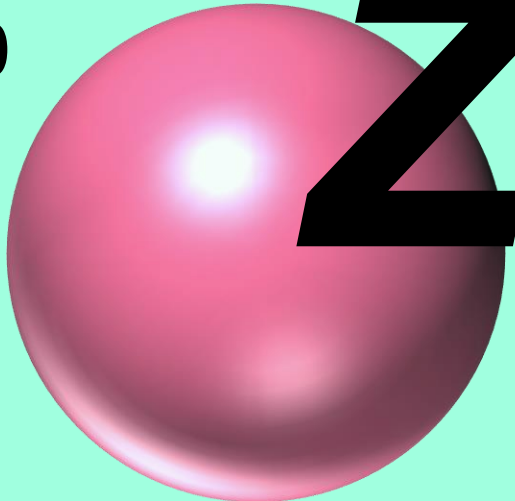
Leptons

Bosons

0
γ
Photon

0
g
Gluon

0

H
Higgs Boson

0

Z
Z Boson

± 1

W
W Boson

**Scalar
Boson**

Vector Bosons

Fermions

$+\frac{2}{3}$

u
Up

$-\frac{1}{3}$

d
Down

-1

e
Electron

0
 ν_e
Electron Neutrino

$+\frac{2}{3}$

c
Charm

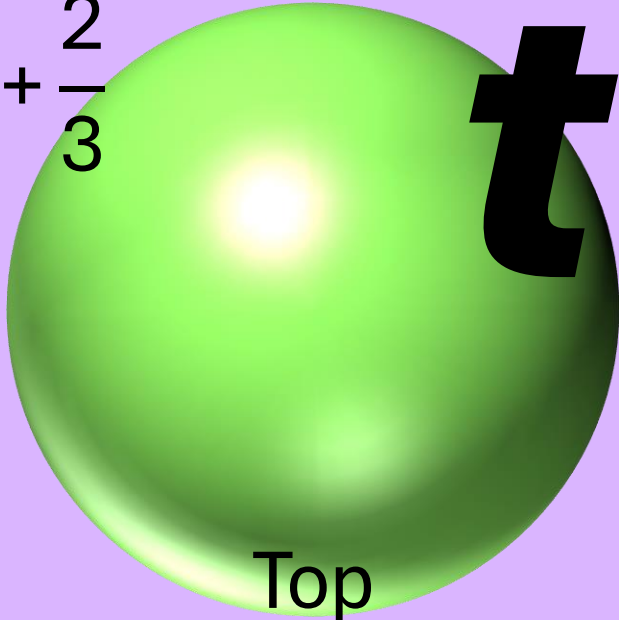
$-\frac{1}{3}$

s
Strange

-1

 μ
Muon

0
 ν_μ
Muon Neutrino

$+\frac{2}{3}$

t
Top

$-\frac{1}{3}$

b
Bottom

-1

 τ
Tau

0
 ν_τ
Tau Neutrino

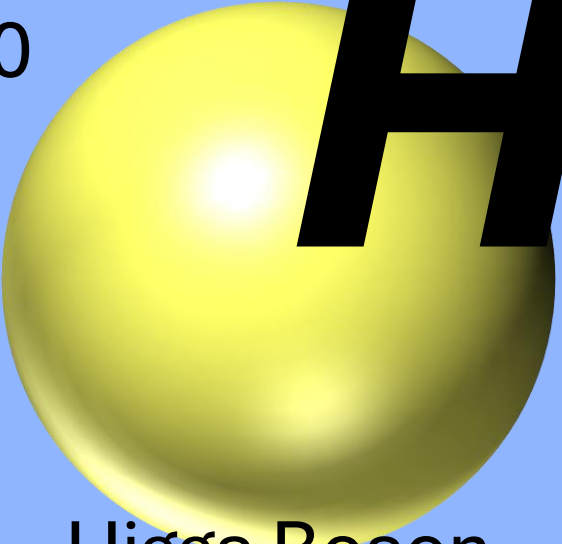
Quarks

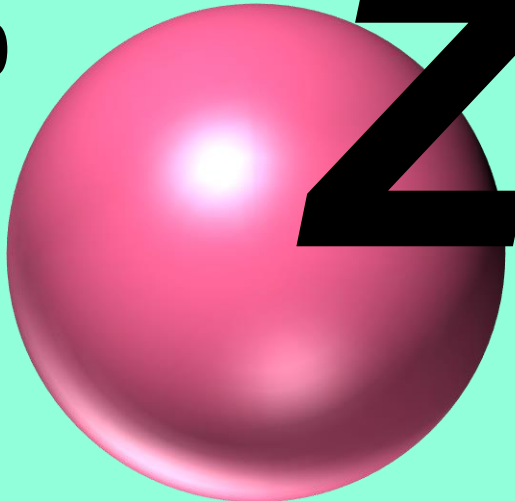
Leptons

Bosons

0
 γ
Photon

0
g
Gluon

0

H
Higgs Boson

0

Z
Z Boson

± 1

W
W Boson

**Scalar
Boson**

Vector Bosons

Fermions

$+\frac{2}{3}$

u
Up

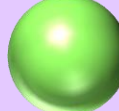
$-\frac{1}{3}$

d
Down

-1

e
Electron

0
 ν_e
Electron Neutrino

$+\frac{2}{3}$

c
Charm

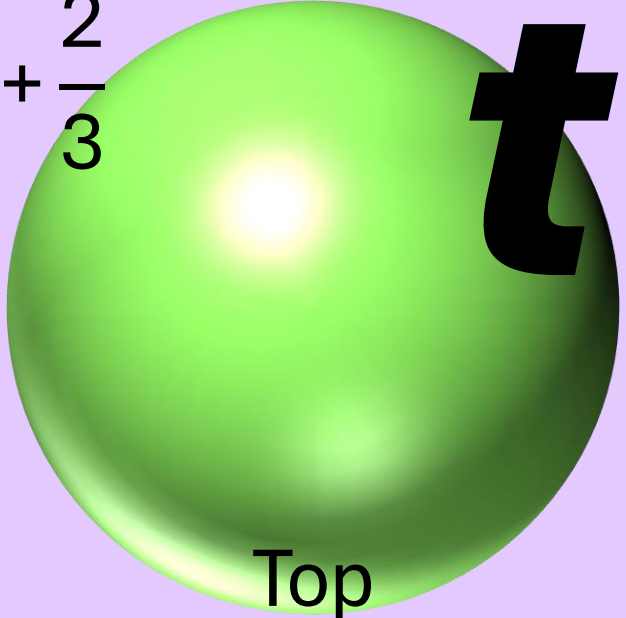
$-\frac{1}{3}$

s
Strange

-1

 μ
Muon

0
 ν_μ
Muon Neutrino

$+\frac{2}{3}$

t
Top

$-\frac{1}{3}$

b
Bottom

-1

 τ
Tau

0
 ν_τ
Tau Neutrino

Quarks

Leptons

Bosons

0
 γ
Photon

0
g
Gluon

0

H
Higgs Boson

0

Z
Z Boson

± 1

W
W Boson

**Scalar
Boson**

Vector Bosons

Fermions

Bosons

POSICÕES MAL

$+\frac{2}{3}$
u

Up

$-\frac{1}{3}$
d

Down

-1
e

Electron

0
 ν_e

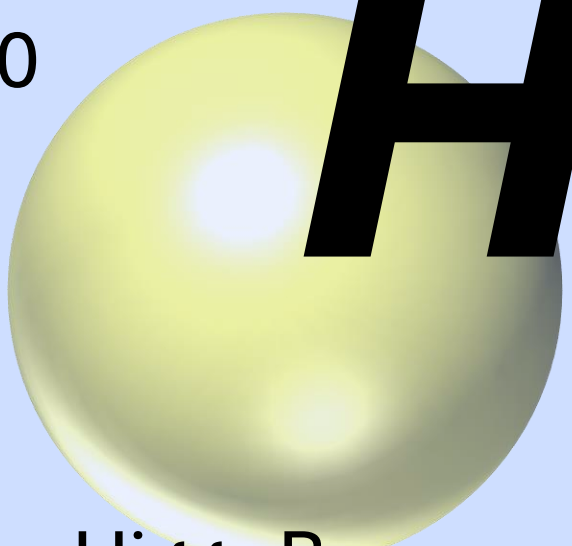
Electron Neutrino

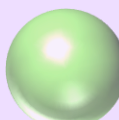
0
 γ

Photon

0
g

Gluon

0
H

Higgs Boson

$+\frac{2}{3}$
c

Charm

$-\frac{1}{3}$
s

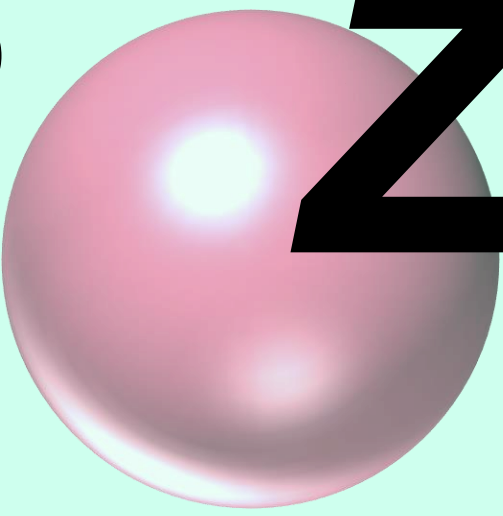
Strange

-1
 μ

Muon

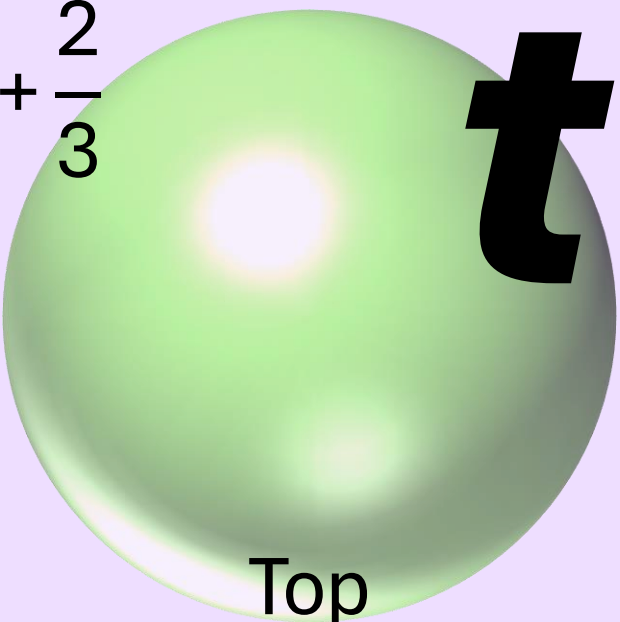
0
 ν_μ

Muon Neutrino

0
Z

Z Boson

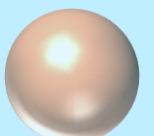
± 1
W

W Boson

$+\frac{2}{3}$
t

Top

$-\frac{1}{3}$
b

Bottom

-1
 τ

Tau

0
 ν_τ

Tau Neutrino

Quarks

Leptons

Vector Bosons

Scalar Boson

Fermions

Bosons

POSICÕES MAL

$+\frac{2}{3}$

u
Up

$-\frac{1}{3}$

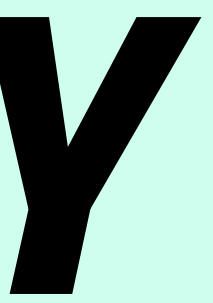
d
Down

-1

e
Electron

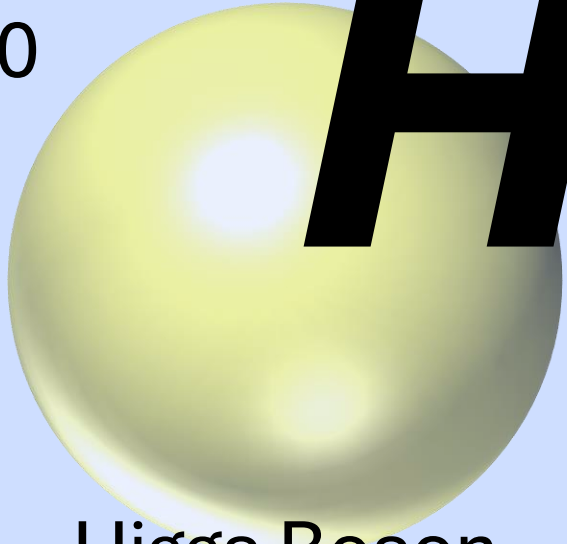
0

 ν_e
Electron Neutrino

0

 γ
Photon

0

g
Gluon

0

H
Higgs Boson

$+\frac{2}{3}$

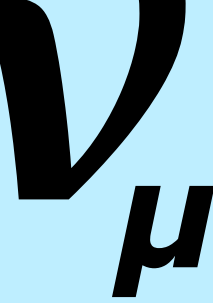
c
Charm

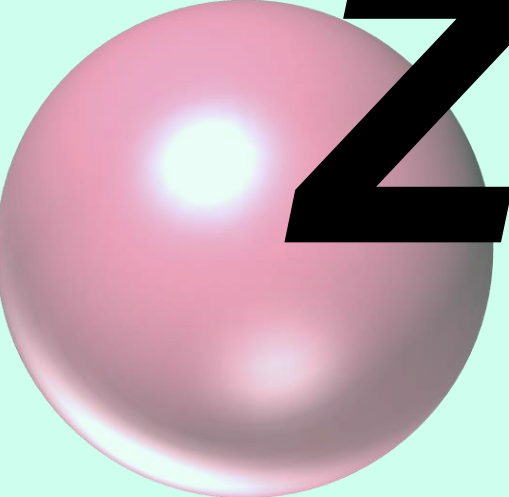
$-\frac{1}{3}$

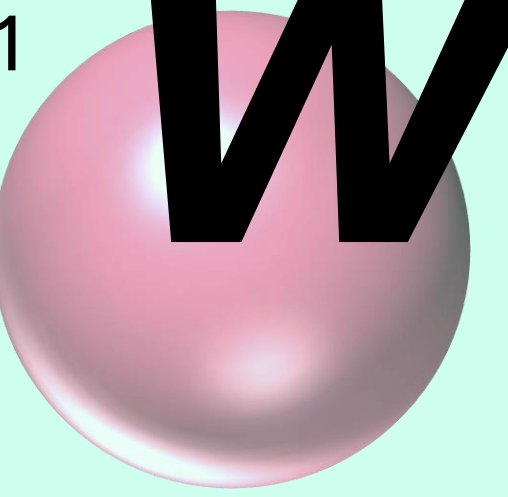
s
Strange

-1

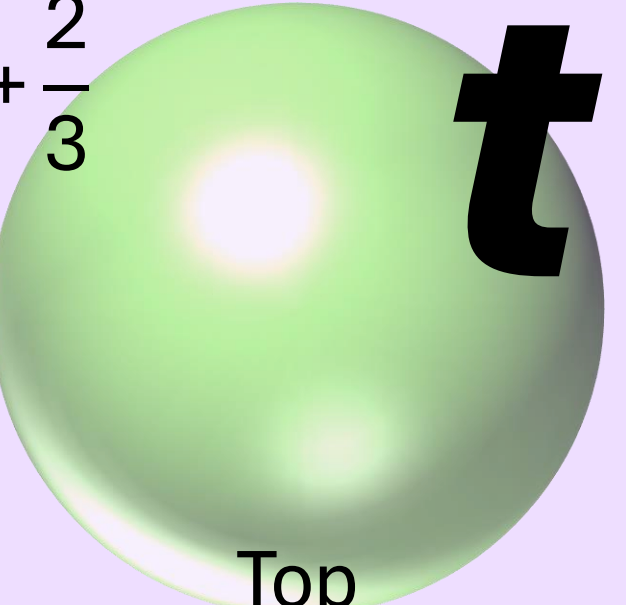
 μ
Muon

0

 ν_μ
Muon Neutrino

0

Z
Z Boson

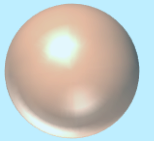
± 1

W
W Boson

Scalar Boson

$+\frac{2}{3}$

t
Top

$-\frac{1}{3}$

b
Bottom

-1

 τ
Tau

0

 ν_τ
Tau Neutrino

Vector Bosons

Quarks

Leptons